Manisha Kasireddy

Data & AI Engineer | M.Sc. Electrical Systems Engineering

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Professional Summary

Azure Data Engineer with three years at Capgemini and eighteen months as a working student at dSPACE GmbH, now completing an M.Sc. in Electrical Systems Engineering at Universität Paderborn with research on LLM-driven reinforcement learning for drone navigation. Microsoft-certified (DP-203, AZ-900); strong in Azure, Databricks, PySpark and SQL, with hands-on LangGraph and RL research experience.

Professional Experience

Research Assistant (WHK), Teaching Support | Universität Paderborn 03/2026 – Present
Course: Electromagnetic Waves and Waveguides (Prof. Dr. J. Förstner) Paderborn, Germany

- Assessed weekly problem sheets against criteria-based rubrics, keeping marking consistent and transparent across the cohort.
- Verified analytical derivations and MATLAB submissions independently before awarding marks, catching errors overlooked in first-pass grading.
- Resolved grading queries directly with students, providing concise, professional written feedback within the tutorial cycle.

Working Student (Werkstudent), Software Testing & Data Analytics | dSPACE GmbH 10/2023 – 04/2025
Simulation and validation solutions for automotive and aerospace embedded systems Paderborn, Germany

- Designed and executed automated regression and unit tests in C#, NUnit and WPF within Visual Studio, extending coverage of internal test tooling.
- Owned the analytics reporting function independently, building test-coverage and defect-analysis dashboards that informed release decisions.
- Integrated structured test results into internal BI workflows and collaborated with developers and QA in Agile sprints, cutting manual reporting effort.

Azure Data Engineer | Capgemini 03/2020 – 03/2023
Enterprise cloud data platforms for international clients Bangalore, India

- Designed and delivered end-to-end Azure data platforms, contributing to architecture decisions on storage layers, orchestration strategy and data modelling.
- Developed metadata-driven ETL/ELT pipelines in Azure Data Factory and Databricks, ingesting structured and semi-structured data (CSV, JSON, Parquet) from REST APIs and CRM systems at scale.
- Designed data warehouse structures using star and snowflake schemas, and tuned T-SQL queries via indexing, partitioning and execution-plan analysis.
- Used PySpark and Python for exploratory data analysis and transformation logic, and delivered Power BI dashboards on operational KPIs for client stakeholders.
- Managed CI/CD releases via Git and Azure DevOps, and right-sized Databricks clusters and Azure storage to control cloud spend.

Summer Intern | Bharat Heavy Electricals Limited (BHEL) 06/2018 – 08/2018
Power generation and heavy electrical equipment manufacturer Visakhapatnam, India

- Supported assembly, testing and quality control of control equipment, including traction, excitation and generator-control systems.
- Gained hands-on exposure to semiconductor and photovoltaic manufacturing processes on the factory floor.

Education

M.Sc. Electrical Systems Engineering | Universität Paderborn, Germany 04/2023 – Present
Focus: machine learning, reinforcement learning, photonic computing

B.Tech. Electronics & Communication Engineering | KIET, Kakinada, India 04/2015 – 06/2019
Thesis: VLSI design of a fixed-width Booth multiplier with an MLCP-based truncation-error estimator

Projects

LRS-Nav: LLM-Guided Reward Generation for Autonomous Drone Navigation 04/2026 – Present
Master's research project, RAT Lab, Universität Paderborn

- Built a LangGraph multi-agent pipeline (Planner, Coder, Reviewer, Tester, Critic) that generates and refines quadcopter reward functions for NVIDIA Isaac Lab, validated automatically through PPO training runs.
- Served a Qwen3 4B model with vLLM and implemented AST-based code injection, Gym environment registration and TensorBoard metric extraction to close the generate–train–evaluate loop.
- Fine-tuned Qwen2.5-Coder-3B with LoRA and DPO in an earlier iteration, raising reward margins from 0.4 to 6.2 and preference accuracy to 1.0 within roughly 60 training steps.

Diffractive Deep Neural Network (D^2NN): Photonic Machine Learning 10/2025 – 03/2026
Research project, Universität Paderborn

- Implemented an all-optical MNIST classifier in TensorFlow/Keras with trainable phase-mask layers and Rayleigh–Sommerfeld free-space propagation, reaching approximately 79% validation accuracy.
- Benchmarked architecture variants: a two-mask design improved accuracy by 11.7% at 1.52× training cost, and a 60λ propagation distance outperformed the baseline.

Technical Skills

Programming Languages: Python, C#, Scala, T-SQL, C++, MATLAB, Bash
Data Engineering: Azure Data Factory, Databricks, PySpark, Spark SQL, ETL/ELT, metadata-driven pipelines, star/snowflake modelling, Microsoft Fabric, Power BI
AI/ML: PyTorch, TensorFlow/Keras, Hugging Face, LangGraph, LLM fine-tuning (LoRA/PEFT, DPO, GRPO), reinforcement learning (PPO, DQN), NVIDIA Isaac Lab, vLLM, FastAPI
Cloud: Microsoft Azure (Data Factory, Databricks, Data Lake Storage, DevOps); HPC clusters (Noctua 2, PC^2)
Databases: Microsoft SQL Server, Azure SQL Database, Azure Data Lake Storage
Tools & Version Control: Visual Studio, VS Code, Jupyter, NUnit, TensorBoard, Streamlit, Conda, Git, GitHub, Azure Repos
Operating Systems: Linux (Ubuntu), Windows

Certifications

Microsoft Certified: Azure Data Engineer Associate (DP-203); Azure Fundamentals (AZ-900)

Languages

English – C2 (proficient) | German – A2/B1 (actively improving) | Telugu – native